Tyler **Kuipers**

Software Architect

Edmonton, Canada

(587) 879-4626 | tkuipers123@gmail.com | tkuipers.ca | tkuipers | tyler-kuipers

About Me

I've been writing software for about 10 years, and the last three I've spent as a Software Architect owning services end to end: design, build, ship, and keeping them healthy after. I led a 2-3 engineer rewrite of a brittle legacy C# system into containerized .NET microservices, with Datadog wired in so reliability and performance were measurable instead of guessed at. I write the design docs and RFCs that get a team aligned on the hard calls, including a feature-flags proposal to decouple deploys from releases. I use AI coding tools every day and have real opinions on where they help and where they get in the way. I work in C#, Python, and TypeScript, lean on Postgres, and I'm comfortable in a modern cloud with Terraform. Mostly I like taking a messy, important domain and leaving it more reliable and pleasant to work in than I found it.

Work Experience

Alberta Motor Association — *Edmonton, Alberta*

Software Architect — *Jan 2023 - Present*

- Led a 2-3 engineer rewrite of a brittle legacy C# document generation system into containerized .NET microservices. 4-5 services (with sidecars) on a managed Kubernetes cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time, replacing a system that had frequent outages. (C#/.NET, Kubernetes, Docker, Helm, Datadog)
- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, REST, OpenAPI, TypeScript)
- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, AWS Lambda, Terraform, TypeScript)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Authored an RFC moving the team to trunk-based development with feature flags. Worked in cross-functional pods alongside PM, design, and QA. Member of the Architecture Steering Committee, and mentor newer developers through legacy codebases and TDD.
- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

Arterys Inc. (Acquired by Tempus Labs) — *Calgary, Alberta (Remote)*

Software Developer — *Aug 2021 - Aug 2022*

- Built and maintained backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association — *Edmonton, Alberta*

Software Developer | Software Team Lead — *Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021*

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions — *Edmonton, Alberta*

Software Developer — *Jan 2018 - Jun 2018*

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada — *Halifax, Nova Scotia*

Associate Software Engineer — *Jan 2017 - Jan 2018*

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

The King's University — *Edmonton, Alberta*

SOFTWARE DEVELOPER/RESEARCH ASSISTANT — *May 2014 - Sep 2016*

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University — *Milton Keynes, England (Remote)*

MSc. COMPUTING (SOFTWARE ENGINEERING) — *May 2024 - Ongoing*

Currently pursuing Master's in Computing at The Open University.

The King's University — *Edmonton, Alberta*

B.S. IN COMPUTER SCIENCE — *Sep 2012 - Dec 2016*

Skills

.NET Services & Ownership — I led a 2-3 engineer rewrite of a brittle legacy C# system into containerized .NET microservices, taking it from frequent outages to something measurable and maintainable. I like owning a service end to end: design doc, build, ship, and the part afterward where it has to stay healthy. I work in C#, Python, and TypeScript, and I care most about leaving a domain more reliable and pleasant to work in than I found it.

Reliability & Observability — I instrument what I ship with Datadog metrics for failure rate, throughput, and latency, and use them to set clear bars for what healthy looks like. I like chasing hard production failures to root cause: a cancellation bug nobody could reproduce got fixed in days once I could replay real production data against it. To me reliability is part of the feature, not a follow-up.

Technical Leadership & Design Docs — I write the design docs and RFCs that align a pod on a hard call, including an RFC to move my team to trunk-based development with feature flags so we could decouple deploys from releases. I set standards through code reviews, PR checklists, and API design guides, and I sit on AMA's Architecture Steering Committee. Making the engineers around me better through clear writing and useful reviews is a real part of how I work.

AI-Forward Development — I use AI coding assistants and agentic workflows every day, and I've formed real opinions on where they speed things up and where they quietly create work. I'm comfortable folding them into the development lifecycle and helping a team get good at using them well, with guardrails rather than blind trust.

Technologies

Backend Languages	C# / .NET, Python, TypeScript, Java, Node.js, Bash
Architecture & APIs	RESTful APIs, OpenAPI, Microservices, Event-Driven Architecture, API Versioning
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Azure DevOps, Azure Artifacts
Databases	PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, GitOps, Feature Flags, Infrastructure as Code
AI-Assisted Development	Claude Code, Cursor, GitHub Copilot, Agentic Workflows